

EEET2370/EEET2371

Assessment Task 3: Project Assignment (30%)

Project Assignment Instructions

- **Start working on this assignment as soon as possible !**
- Check canvas for the due date of this assignment (Research Project)
- This written assignment (Research Project) worth **30%** of the total mark
- This assignment is **individual**
- This assignment gives a large room for the student to be **innovative and to propose their own ideas**
- Marks will be awarded as follows (details is in the rubric on Canvas)
 - Depth of literature review
 - Figures, plots and illustrations (drawn by the student)
 - Clarity of writing and equations
 - Novelty and innovation
 - Sound assumptions and parameters
 - Simulation code
 - Proper references
- The assignment should be written in LaTeX format using IEEE Conference Template. It is suggested to use the online Overleaf editor on: <https://www.overleaf.com/project>. Feel free to use any other LaTeX editor (online or offline).
- Use IEEE citation style
- It is expected to have a document of 4-6 pages long - double column format (**3000-4500 words**) – excluding the code which should be submitted separately
- The student is expected to develop a code to simulate/analyze the selected concept, the code needs to be neatly written with **clear annotation (comments)**. Also, it needs to be consistent with the assignment text and well explained.
 - The code needs to be (written in ASCII format) with sufficient annotation to allow the examiner to understand the code.
 - Coding/scripting language needs to be either in **Matlab (or Simulink), Python, or C/C++**
 - It is indeed allowed to use libraries and functions (both built-in and external), however it should be clearly identified in the code which part was developed by the student and which part was taken from other sources.
 - The code contribution of the student should be significant and not just adjusting available online codes
- **State your assumptions inside the assignment document! Marks are awarded to sound assumptions**
- Important note about **plagiarism**:
 - You should write the assignment yourself. Under no circumstances any part of this assignment can be written by anyone else (unless otherwise some quoted text is clearly stated and referenced)
 - Plagiarism will not be tolerated and heavy loss of marks (or **zero mark**) will be incurred with possible wider academic penalties.
 - The rules apply similarly to the developed codes



Report structure

Following is the recommended report structure,

- **Abstract** (min 150 words)
 - Provide an interesting motivation for the reader to read the rest of the report !
 - Include a quick background/ motivation
 - Include a summary of the main contributions of your assignment
 - Include the main take-away results
- **Introduction and Background** (min 400 words)
 - Background on your topic
 - Motivation for studying your topic
 - Main contribution of your work (dot points) and how it is different from the literature (novelty)
- **Literature review** (min 1000 words)
 - Literature review on the general subject
 - Literature review on your particular topic
 - (if applicable) Comparison between the literature qualitatively and/ or quantitatively
 - Include comparison tables when applicable
- **System model***
 - Feel free to call this section something else (for example network model, simulator model, ...) or to split this section into multiple sections / subsections
 - Also feel free to take some of these points to the next section when needed
 - Main assumptions for your work
 - Utilized models (examples – when applicable: Channel models, fading, packet error rate, error models, etc..)
 - Include explanation / justification on why did you pick up this model / modelling approach
 - A diagram for your system (or diagrams) – neatly drawn using professional tools such as Visio, Power point, etc..
 - Equations and derivation
 - Analytic results when applicable
 - Add graphs, plots
 - Explain your code / algorithms
- **Results and performance***
 - Feel free to call this section something else or to split it into multiple sections / subsections
 - Simulation numerical inputs and parameters (why did you pick up these parameters, add references)
 - Figures graphs of your results, properly annotated with legends, units, clear x and y axis ticks, etc..
 - (bonus) to compare your results with existing literature
- **Conclusion** (min 150 words)
 - Summary of the paper
 - Take away message
 - Future work that can be done on this topic

*System Model and Results sections both minimum 1300 words

Topics

Below are the topics that will be allocated to students,

Topic 1 Radio coverage designs for LPWAN access network (undergraduates)

The objective is to design radio coverage ([downlink](#)) for a **NB-IoT access network** to cover Melbourne metropolitan area. The project will use existing cellular tower locations as provided on Canvas (“TowersLocations.csv”, assume the tower height as 30 m). The main **outcomes** of the project are:

- The project will provide literature review for (but not limited to):
 - o Brief overview of NB-IoT standard
 - o NB-IoT coverage and performance
 - o NB-IoT limitations
 - o State-of-the-art coverage design approaches for NB-IoT (or in Cellular coverage in general)
- Map(s) showing the locations of the import sites locations
- Graph(s) for the Block error rate (BLER) vs. SNR (for two repetition rates 1 and 32)
- Graph (s) for the radio coverage of NB-IoT (in terms of the received signal strength – dBm)
- Graph (s) for the radio coverage of NB-IoT (in terms of the signal to noise ratio)
- Two plots for coverage contour for two different repetition rates 1 and 32. Take 5% BLER as the threshold
- Additional figures and results as needed

Hints:

- In order to get more realistic coverage, assume an average background interference power = -85 dBm and use this as equivalent to noise, i.e. Effective noise power = $N_{\text{thermal}} + \text{average interference power}$. **Make sure to have this summation in linear**
- You could make use of Matlab NB-IoT package to extract the block error rate (BLER) vs. SNR and utilize this for mapping the SNR into coverage. [See this tutorial](#).
- You could make use of Matlab “RF Propagation” toolbox. [See this example](#).
- You could make use of [geoplot](#) function in Matlab to show the location of the towers
- See 3GPP recommendations for NB-IoT design parameters (noise floor
- Use transmit power of 40 watts with 3 directional antennas (120° spaced) at each site (each having gain of 12 dBi) – You could use the antenna pattern uploaded in Canvas ‘Directional12dBi.mat’, or create your own antenna pattern.
- Use frequency 900 MHz for the simulation

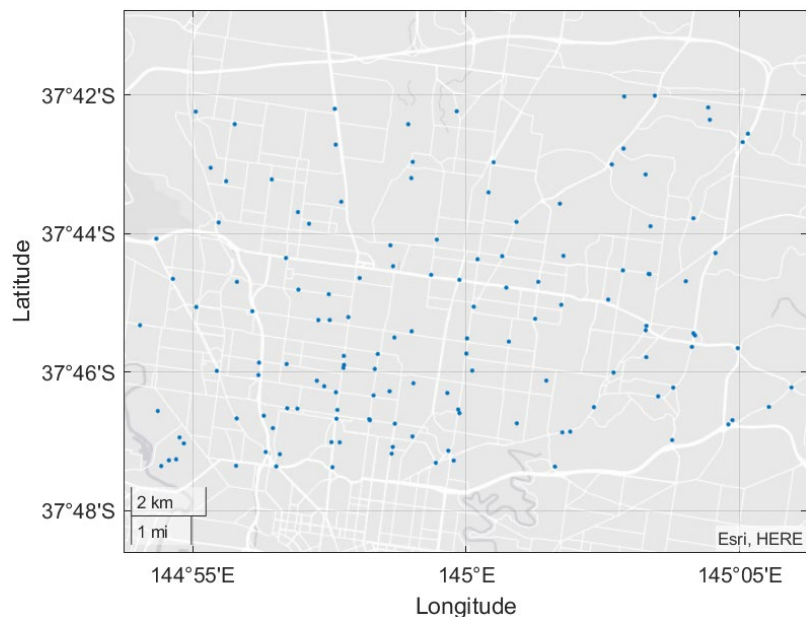


Figure 1 Tower locations on Canvas, file name: "TowersLocations.csv"

- Use Longley-Rice model as your radio propagation model

Recommended reading on a [similar practical project](#) in Victoria conducted by RMIT University team and 5 Victorian councils.

If something is not stated in the question **use your own assumptions** and explain the reason for such selection.

Topic 2 IoT (LPWAN) access network over UAVs (Post-graduates)

This project is similar to “Topic 1” where the objective is to simulate the downlink radio coverage for an **NB-IoT** eNB (base station) fitted onboard of a flying UAV. The location of the UAV is over Melbourne CBD, hovering above the intersection of Bourke Street and Elizabeth Street. The project will make use of the open street map file on canvas “melbourne.osm” that includes the buildings’ height as shown in the figure below. The main **outcomes** of the project:

- The project will provide literature review for (but not limited to):
 - o Brief overview of NB-IoT standard
 - o NB-IoT coverage and performance
 - o Using UAV for providing cellular coverage (what are the pros / cons of UAVs)
 - o State-of-the-art coverage design approaches for NB-IoT (or in Cellular coverage in general)
- Map(s) showing the buildings and the UAV (as points at different heights 30 m , 90 m, 120 m, 150 m, 250m)
- Graph(s) for the Block error rate (BLER) vs. SNR (for two repetition rates 1 and 32)
- Graph(s) for the radio coverage of NB-IoT (in terms of the received signal strength – dBm) for heights 30 m , 90 m, 120 m, 150 m, 250 m.
- Graph(s) for the radio coverage of NB-IoT (in terms of the signal to noise ratio) for the specified heights
- Coverage contour plot for two different repetition rates 1 and 32. Take 5% BLER as the threshold (use 120m altitude)
- Additional figures and results as needed

Hints:

- In order to get more realistic coverage, assume an average background interference power = -85 dBm and use this as equivalent to noise, i.e. Effective noise power = $N_{\text{thermal}} + \text{average interference power}$. **Make sure to have this summation in linear**
- You could make use of Matlab NB-IoT package to extract the block error rate (BLER) vs. SNR and utilize this for mapping the SNR into coverage. [See this tutorial](#).



Figure 2 Melbourne CBD buildings as provided in the file "melbourne.osm"

- You could make use of Matlab “RF Propagation” toolbox, ([the ray tracing feature](#))
- You could make use of `siteviewer` function in Matlab to show the location of the towers
- See 3GPP recommendations for NB-IoT design parameters (for the noise floor)
- Use Tx power of 10 watts at the UAV with an isotropic antenna gain 0 dBi
- Use frequency 2100 MHz for the simulation

Recommended reading on a [similar practical project](#) in Victoria conducted by RMIT University team and 5 Victorian councils.

If something is not stated in the question **use your own assumptions** and explain the reason for such selection